

A four-vertex counterexample to Knudson's persistence-gradient conjecture

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Abstract

Knudson conjectured that every nonincident persistence pair is joined by exactly one gradient path in the discrete vector field obtained by retaining the incident persistence pairs. We give a simplex-wise filtration of a triangulated interval with four vertices and three edges for which one such pair has no gradient path. This disproves Conjecture 2 in his contribution to Oberwolfach Report 29/2008. The proof consists of three boundary reductions and an inspection of the matching. The same example works over any field.

1 The statement and the example

Let P be the persistence pairing of a finite simplicial complex whose filtration adds one simplex at a time, with coefficients in $\mathbb{F}_2$. For a codimension-one face relation $\sigma < \tau$, put

$$V_P = \{(\sigma, \tau) \in P : \sigma < \tau\}.$$

Knudson's Conjecture 2 [1, p. 1629] predicts a unique gradient path in this fixed field for every nonincident pair in P . We use the directed incidence graph: unmatched incidences point from a simplex to a facet, and matched incidences point upward. A simplex is critical if it belongs to no matching pair. A gradient path from a critical edge begins by descending to an endpoint. Each subsequent upward step uses a matched incidence and is followed by descent to the other endpoint of that edge.

Take vertices a, b, c, d and edges cd, bd, ac , with no higher simplices. Here xy denotes $\{x, y\}$. The underlying space is the interval $a-c-d-b$. Insert the seven nonempty simplices in the order

$$a, b, c, d, cd, bd, ac. \tag{1}$$

Every face appears before its coface, so this is an admissible filtration. The indices $1, \dots, 7$ order the filtration; the field V_P is constructed from the persistence pairs.

Theorem 1. *For the filtration (1), the finite persistence pairs and their incident submatching are*

$$P = \{(d, cd), (c, bd), (b, ac)\}, \quad V_P = \{(d, cd)\}.$$

The pair (b, ac) is nonincident, and there are zero gradient paths from ac to b in V_P . Consequently Knudson's Conjecture 2 is false.

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Proof. Order the vertex rows as $a < b < c < d$ and edge columns as $cd < bd < ac$. Over $\mathbb{F}_2$, the usual left-to-right persistence reduction [2] eliminates repeated latest nonzero rows using earlier reduced columns:

$$\begin{aligned} R(cd) &= \partial(cd) = c + d, \\ R(bd) &= \partial(bd) + R(cd) = (b + d) + (c + d) = b + c, \\ R(ac) &= \partial(ac) + R(bd) = (a + c) + (b + c) = a + b. \end{aligned}$$

The three pivots are respectively d, c, b , giving precisely the stated pairs. The vertex a is unpaired. Of those pairs, only $d < cd$ is an incidence, giving the stated V_P .

For an independent check of the pairing, cd kills the component born at d ; bd merges the component born at c into the older one born at b ; and ac merges that surviving component into the oldest one, born at a . This is the same pairing in ordinary H_0 .

The complete directed incidence graph has the arrows

$$ac \rightarrow a, \quad ac \rightarrow c, \quad bd \rightarrow b, \quad bd \rightarrow d, \quad d \rightarrow cd, \quad cd \rightarrow c.$$

It is acyclic. In particular V_P is a discrete gradient, and both b and ac are critical. Either first step from ac reaches a or c ; both are unmatched vertices, so the path stops. The reachable set is exactly $\{ac, a, c\}$, which excludes b . $\square$

2 Why cancellation does not repair the fixed field

The reduction of ac uses $R(bd)$ to remove its pivot at c . But the persistence pair (c, bd) is nonincident and contributes no upward step to V_P . Algebraic column elimination therefore need not trace a path in that field.

There is a unique path for the earlier pair (c, bd) :

$$bd \rightarrow d \rightarrow cd \rightarrow c.$$

Reversing it gives the new matching $V' = \{(d, bd), (c, cd)\}$, in which the path

$$ac \rightarrow c \rightarrow cd \rightarrow d \rightarrow bd \rightarrow b$$

now exists. This observation concerns V' , whereas the conjecture specifies V_P . It is consistent with the role of ordered cancellations in the surface simplification results of Bauer, Lange, and Wardetzky [3, Lemma 9]; we claim no counterexample to those results.

Other coefficients. Orient each edge from its alphabetically earlier to its later vertex. Over any field the identical elimination, with subtraction, gives

$$R(cd) = d - c, \quad R(bd) = (d - b) - (d - c) = c - b, \quad R(ac) = (c - a) - (c - b) = b - a.$$

All three pivot coefficients are 1. The pairs, matching, and absence of a path are unchanged. No minimality claim is needed or made.

Reproducibility and status. The [companion package and audit record](#) provide two standard-library Python verifiers: one uses boundary reduction and union-find; the other reconstructs the H_0 rank invariant from component partitions and also checks signed reduction and path reversal. The proof above is independent of these programs. A documented literature audit completed on 23 September 2026 located no earlier refutation of the stated conjecture. This is a bounded search, not a guarantee of priority; in particular, an [author bibliography](#) lists related work in preparation. The audit records inaccessible sources. This unrefereed note and its checks were prepared with OpenAI Codex and separate AI audit agents, from a candidate construction supplied by the author from prior AI-assisted work.

References

- [1] K. P. Knudson, Persistent homology and discrete Morse theory, in *Computational Algebraic Topology*, Oberwolfach Report 29/2008, *Oberwolfach Reports* **5** (2008), 1628–1630. Conjecture 2, p. 1629. [doi:10.4171/OWR/2008/29](https://doi.org/10.4171/OWR/2008/29).
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